

Human-Computer Interaction

Capturing and utilizing human silhouette in interface design

- The aim of the classes is to familiarize with the capabilities of modern human skeleton detection systems based on camera images.
- We will focus on the `MediaPipe` library and its models.

Exercise 1 – Working with camera in Python

- Execute and analyse the below code for image acquisition using the `OpenCV` library.

The code is provided in a separate file attached to this instruction

```
1  import cv2
2
3  # Initialize the camera
4  cap = cv2.VideoCapture(0) # Camera number (0 for the default camera)
5
6  while True:
7      ret, frame = cap.read()
8      if not ret:
9          continue
10
11     # Mirror the image horizontally
12     frame = cv2.flip(frame, 1)
13
14     # Display the image
15     cv2.imshow("Camera image", frame)
16
17     # Exit the program when the 'q' key is pressed
18     if cv2.waitKey(1) & 0xFF == ord('q'):
19         break
20
21 cap.release()
22 cv2.destroyAllWindows()
```

Exercise 2 – Working with MediaPipe Hand model in Python

- Extend the code from Exercise 1 to detect hand landmarks. Use the following steps.
- Import the `MediaPipe` library
 - `import mediapipe as mp`
- Initialize the `MediaPipe Hands` object
 - `mp_hands = mp.solutions.hands`
 - `hands = mp_hands.Hands()`
- Initialize the drawing utility
 - `mp_drawing = mp.solutions.drawing_utils`

- Process each frame to detect and draw hands

Remember of the BGR color space convention (used in OpenCV)

```
o frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
```

To detect hands in the image use:

```
o results = hands.process(frame_rgb)
```

Draw all landmarks and hand skeleton connections on the image

```
o if results.multi_hand_landmarks:
o     for hand_landmarks in results.multi_hand_landmarks:
o         mp_drawing.draw_landmarks(frame, hand_landmarks,
                                     mp_hands.HAND_CONNECTIONS)
```

Exercise 3 – Working with MediaPipe Face Mesh in Python

- Based on previous exercises, write code that captures an image from the camera and performs face detection and landmark detection using the Face Mesh model from MediaPipe.

Exercise 4 – Working with MediaPipe silhouette model in Python

- Change the model to a holistic one from MediaPipe, which computes the detection of 33 topological points of the human silhouette.
- Test the model's performance.
- What are its strengths and weaknesses?

Report

- Present and discuss your results with the group.
- Discuss potential applications in touchless human-machine interaction.
- Code submission is not required for this exercise. The topic will be expanded in subsequent classes.